

How SSL/TLS Works — Simple Explanation

SSL (technically now called TLS) encrypts the connection between a browser and a server so no one in between can read or tamper with the data.

Working of the ssl certificate

- ❓ **Browser connects** to your server on port 443 (HTTPS)
- ❓ **Server sends its certificate** — this proves "I am really yourdomain.com" and includes a public key
- ❓ **Browser verifies the certificate** — checks it was issued by a trusted Certificate Authority (like Let's Encrypt) and hasn't expired
- ❓ **Browser and server agree on a shared secret key** using the public/private key pair (this is called the key exchange)
- ❓ **All further data is encrypted** using that shared secret — this is fast symmetric encryption, not the slower public-key kind

Commands for setting HTTPS in nginx

1. Install Nginx (if not already installed)

```
#sudo apt update
```

```
#sudo apt install nginx -y
```

2. Allow HTTP/HTTPS through firewall

```
#sudo ufw allow 'Nginx Full'
```

3. Install Certbot with Nginx plugin

```
#sudo apt install certbot python3-certbot-nginx -y
```

4. Make sure your domain's DNS A record points to your server's IP

5. Get and auto-install the SSL certificate

```
#sudo certbot --nginx -d yourdomain.com -d www.yourdomain.com
```

- It will ask for your email (for renewal alerts)
- Agree to terms
- It automatically edits your Nginx config and sets up HTTP → HTTPS redirect

6. Test Nginx config

```
#sudo nginx -t
```

7. Reload Nginx

```
#sudo systemctl reload nginx
```

8. Test auto-renewal (certs expire every 90 days)

```
#sudo certbot renew --dry-run
```

Config File Location (auto-created by Certbot)

```
/etc/nginx/sites-available/yourdomain.com
```

Certificate File Location

```
/etc/letsencrypt/live/yourdomain.com/fullchain.pem
```

```
/etc/letsencrypt/live/yourdomain.com/privkey.pem
```